Digital Filtering Techniques in SDR Pipelines: Audio Conditioning and PSD Smoothing

1. Introduction

Digital Signal Processing (DSP) is the core of Software Defined Radio. Once a signal is digitized and down-converted, filtering becomes the primary tool for extracting information and reducing noise. This report details two critical software-defined filtering applications: post-demodulation audio conditioning and Power Spectral Density (PSD) stabilization for spectrum sensing.

2. Post-Demodulation Audio Filtering

The objective of this stage is to refine the output of AM/FM/SSB demodulators by removing out-of-band noise and improving the Signal-to-Noise Ratio (SNR) for the end-user or automated recognition systems.

2.1 Mathematical Framework

After demodulation, the signal $x[n]$ often contains high-frequency artifacts and residual carrier noise. We apply a Linear Time-Invariant (LTI) system, typically a Low-Pass Filter (LPF), defined by the convolution:

$$y[n] = \sum_{k=0}^{M-1} h[k]\, x[n-k]$$

Where $h[k]$ represents the impulse response of the filter. For voice communication, the cutoff frequency f_c is typically set between 3 kHz and 5 kHz.

2.2 FIR vs. IIR Implementations

- **Finite Impulse Response (FIR):** Preferred for SDR due to inherent stability and linear phase response, which prevents group delay distortion. Common windowing methods (Hamming, Blackman) are used to minimize Gibbs phenomenon ripples.
- **Infinite Impulse Response (IIR):** Offers steeper roll-off with fewer coefficients (lower computational cost) but may introduce phase non-linearity and potential instability in fixed-point implementations.

2.3 Advanced Research: De-emphasis and Noise Blanking

In FM systems, high-frequency noise increases linearly. Research suggests that a de-emphasis filter (a specialized LPF with a specific time constant, e.g., $75\,\mu$s) is mandatory to restore the original audio balance. Furthermore, integrating a *Noise Blanker* before the filter can effectively suppress impulsive noise (spikes) before they are "smeared" by the filter's time constant.

3. Spectral Smoothing for PSD Estimation

Raw Power Spectral Density (PSD) estimates, such as those derived from a single Periodogram, exhibit high variance, making it difficult to distinguish weak signals from noise floor fluctuations.

3.1 The Moving Average Filter

To stabilize the spectrum for sensing and visualization (waterfall plots), a sliding window average is applied across the frequency bins k:

$$\tilde{P}[k] = \frac{1}{N} \sum_{i=0}^{N-1} P[k-i]$$

Where N is the window size. This operation acts as a low-pass filter on the spectral representation itself.

3.2 The Variance-Resolution Trade-off

The selection of N is a critical design decision:

- **Large N:** Significantly reduces variance and stabilizes the noise floor, which is ideal for detecting wideband occupancy or energy-based sensing. However, it leads to *spectral blurring*, where narrow-band carriers (e.g., CW or narrowband IoT) may be suppressed or merged.
- **Small N:** Preserves frequency resolution and identifies narrow peaks but leaves the PSD "jagged," increasing the probability of false alarms in automated threshold detectors.

3.3 Advanced Research: Welch's Method and EMA

Current research in spectrum monitoring favors **Welch's Method**, which averages multiple overlapping FFTs in the time domain rather than just smoothing bins in the frequency domain. This reduces variance without the same degree of spectral leakage.

For real-time SDR waterfalls, an **Exponential Moving Average (EMA)** is often implemented to provide temporal persistence:

$$S_t = \alpha \cdot P_t + (1-\alpha) \cdot S_{t-1}$$

Where α determines the "memory" of the spectrum, allowing users to see transient signals (bursts) while maintaining a stable background.

4. Implementation Best Practices

- **Computational Efficiency:** Use SIMD (Single Instruction, Multiple Data) instructions or FFTW libraries for large-scale PSD calculations to maintain real-time throughput.
- **Metadata Integration:** Always store the filter parameters (taps, window type, N) alongside the IQ data to ensure that subsequent analysis can "undo" or account for the filter's bias.
- **Dynamic Thresholding:** Use the smoothed PSD to calculate a *Constant False Alarm Rate* (CFAR) threshold, which adapts to changing noise environments in real-time.